to 70MB large data files!

The demodulator is an excellent diagnostic tool, which can be found very useful when a satellite signal is being received. In the Diagram option, we can choose the input side or output side signal quality as the Constellation or Eye diagrams. The Constellation diagram is an easy diagnostic tool, as can be seen later when actual signal is received, as it shows the four phases (Quadrature Phase Modulation!) and magnitudes in each of the four quadrants.

Step 3: Recording a satellite pass. The final step in the first approach is to record the actual satellite pass. For this, we know the Meteor M2 satellite transmits at 137.100MHz. Find the satellite pass from your Prediction software, tune the receiver to this frequency, and start the demodulator by checking the Demodulator check-box, as shown in Fig. 5(a). As the satellite rises above horizon, the signal strength increases, and you shall see that the Constellation diagram starts changing from random dots to a circular 'O' ring and then finally stabilises with four dots indicating that the 72kHz QPSK signal is being received.

At the same time two important developments take place. First, the demodulator shows a 'Locked' status in red colour and the File Write shows the data being written to the file. Refer Fig. 5(b) and Fig. 5(c). When the satellite leaves your horizon or field of view, the demodulator also loses lock onto the carrier and any data write also ceases. For a good and successful satellite pass, one can expect a file size in excess of 60MB, although smaller file sizes of 35MB are also possible and would result in useful earth imagery.

Once your satellite pass is complete, stop the demodulator and uncheck the Demodulator check-box. The demodulated data file is saved with a '.S' extension in the folder you chose while configuring the demodulator.

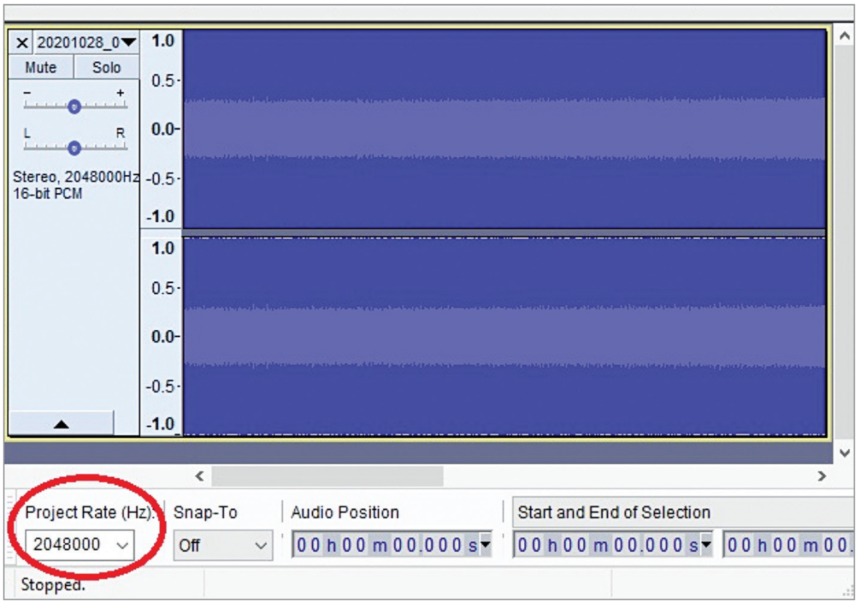

Fig. 7: Resampling the I-Q Wave file using Audacity

Since the image extraction process is common to both approaches, we shall see this process at a later point in the text.

Manual recording using IQ wave file. The other method is to manually record the IQ signal in the form of a .wave file and then demodulate it as part of the post processing. Although this is a very cumbersome and time consuming process, sometimes it proves advantageous.

Step 1: Recording the satellite pass using the I-Q wave file. Whether you are using the RTL-SDR with the SDR# or the RSP1A receiver with SDRUno software, both these receiver GUIs offer you a facility to record the sound or the baseband I-Q (Quadrature Modulation) data as a wave file. Refer Fig. 6 for this. Kindly note that this wave file is the actual RF signal recorded and not a demodulated output.

Step 2: Processing the wave file using Audacity. The next step is to resample the recorded RF signal at a lower sampling rate. This is necessary for the demodulation process in the next step. For this, open the satellite wave file (I-Q data) in Audacity software.

The file might be large, several GB in size! And you can notice the sampling rate at which the SDR recorded this wave file in the bottom left corner of the Audacity GUI. Refer Fig. 7 for details.

Next, change this sampling to 130ksps. For example, if your sampling rate is shown as 2048000, change it to 130000. Then, select the entire waveform/signal and save the File/data as (Save as…) wave file. Once saved properly, you would notice that the file size has reduced from GBs to a few hundred MBs. You can delete the larger files to save space on the hard disk as these are not needed any more.

Step 3: Manually demodulating the wave file. The next step is to demodulate this wave file using the LrptRx Demodulator software, which the author found from the websites mentioned above. The link is: *https://www.dropbox.com/s/qq1fjyitpa3j14o/software.zip*

The dashboard of LRPTRX, as shown in Fig. 8(a), is very complex but we do not have to worry about that. We just need to enter the Input and Output filenames and their